

PANCREAS (Surgery)

Anatomy & physiology

- **Retroperitoneal.** Head = 30% mass; body + tail = 70%. Weighs ~80 g. **80–90% exocrine acinar tissue** (lobules).
- **Islets of Langerhans (endocrine):** **75% B cells (insulin)**, **20% A cells (glucagon)**, rest D cells (somatostatin) + PP cells.

Investigation

- **Serum enzymes:** **amylase = most widely used** (rises in hours, falls over 4–8 days); **lipase = more sensitive + specific** (less available).
- **Causes of raised amylase OTHER than pancreatitis:** upper GI perforation, mesenteric infarction, viscus torsion, retroperitoneal haematoma, ectopic pregnancy, **macroamylasaemia, renal failure, salivary gland inflammation.**
- Morphology: US, CT, MRI, ERCP, EUS, plain X-ray. Normal duct 2mm.

Acute pancreatitis

Definition (2 of 3)

1. **Abdominal pain**, 2. **amylase/lipase $\geq 3\times$ normal**, 3. **characteristic CT findings.**

- **Mechanism:** premature activation of pancreatic enzymes $\rightarrow$ **autodigestion** $\rightarrow$ oedema, haemorrhage, necrosis. Inflammatory mediators $\rightarrow$ systemic complications (haemodynamic instability, bacteraemia, **ARDS**, GI haemorrhage, renal failure, **DIC**).

Categories

- **Mild (interstitial oedematous):** most patients, mortality ~1%.
- **Severe (necrotising):** 5–10%; pancreatic necrosis + severe SIRS + multi-organ failure; mortality 20–50%.

Epidemiology & aetiology

- 3% of abdominal pain admissions. Peak: young men + older women.
- **Biliary calculi 50–70% (commonest), alcohol 25%.**
- **"I GET SMASHED":** Idiopathic, Gallstones, Ethanol, Trauma, Steroids, Mumps/malignancy, Autoimmune, Scorpion sting, Hypercalcaemia/hyperlipidaemia/hereditary, ERCP (post), Drugs (azathioprine, valproate, thiazides, oestrogens). Also pancreas divisum, post-surgery.
- **Establish aetiology; if gallstones $\rightarrow$ cholecystectomy same admission.**

Clinical

- **Pain = cardinal** (rapid, max in minutes, epigastric, **radiates to back ~50%**, refractory to analgesia). N/V/retching. Tachypnoea, tachycardia, hypotension.
- **SIRS = ≥ 2 of:** HR >90 , temp $<36/>38$, RR >20 or pCO₂ <32 , WBC $<4000/>12,000$.
- Mild icterus (gallstone), swinging pyrexia (cholangitis), **Grey Turner's (flank) / Cullen's (umbilicus) sign** (haemorrhage), ileus/ascites, epigastric mass, guarding, **pleural effusion 10–20%.**

Investigations (3 questions: diagnosis? severity? aetiology?)

- **Amylase $\geq 3\times$ normal** diagnostic; normal doesn't exclude (late presentation); **lipase more sensitive/specific. CECT = best single imaging** if doubt.
- **Severity scoring:** at 24h/48h/7d. **Ranson + Glasgow (PANCREAS) — score ≥ 3 at 48h = severe.** Persisting organ failure = severe. **CRP >150 mg/L at 48h = severity.** BMI >30 = higher risk.

- **Glasgow (PANCREAS):** PaO₂ <8kPa, Age >55, Neutrophils WBC >15, Calcium <2.0, Renal urea >16, Enzymes LDH >600, Albumin <32, Sugar glucose >10.
- **Atlanta classification (2013):** **Mild** (no organ failure/complications); **Moderately severe** (transient organ failure <48h ± local complications); **Severe** (persistent organ failure >48h).
- **Imaging:** plain film — **sentinel loop, colon cut-off sign, renal halo**. US within 24h (gallstones, exclude cholecystitis, CBD). **CECT** for: doubt, distinguish interstitial vs necrotising (**Balthazar criteria**; underestimates necrosis first 72h), organ failure/sepsis, local complications. ERCP for gallstone pancreatitis + cholangitis (urgent).

Management

- **Mild** → **conservative**: IV fluids, observation, brief fasting if nauseated, **NO antibiotics**, analgesia/antiemetics; CT only if deteriorating.
- **Severe** → **HDU/ICU**: analgesia, **aggressive fluid resuscitation**, O₂, invasive monitoring, NG drainage (initially), **antibiotics if cholangitis** (prophylactic considered), CT if deteriorating, **ERCP within 72h for severe gallstone pancreatitis/cholangitis**, organ support, **enteral (NG) feeding** if nutrition needed.

Complications

- **Systemic (first week):** shock, arrhythmias, **ARDS**, renal failure, **DIC**, hypocalcaemia, hyperglycaemia, ileus.
- **Local (after first week):**
- **Acute peripancreatic fluid collection (APFC):** early, no wall, sterile, usually resolves.
- **Pancreatic necrosis** (no parenchymal enhancement on CECT) → **acute necrotic collection (ANC)** → after 4 weeks → **walled-off necrosis (WON)**. **Infected necrosis = mortality up to 50%; sterile necrosis should NOT be drained.**
- **Pancreatic abscess** (circumscribed pus).
- **Pseudocyst:** amylase-rich fluid in fibrous/granulation wall; follows mild pancreatitis, outside pancreas (unresolved APFC); **needs ≥4 weeks to form**. Complications: infection (abscess/sepsis), rupture (GI bleed/fistula/peritonitis), enlargement (obstructive jaundice/bowel obstruction/pain), **erosion into vessel** → **haemorrhage**.
- Also: ascites, pleural effusion, portal/splenic vein thrombosis, pseudoaneurysm.

Chronic pancreatitis

- Continuing inflammation → **irreversible morphological change**, pain ± permanent loss of function (may be painless). May be same spectrum as acute.

High-yield one-liners

- **Diagnosis = 2 of 3 (pain, amylase/lipase ≥3×, CT).** Lipase more sensitive/specific than amylase.
- **Gallstones (50–70%) + alcohol (25%) = main causes; I GET SMASHED mnemonic.**
- **Severity: Ranson/Glasgow ≥3 at 48h = severe; CRP >150 at 48h; persisting organ failure.** Atlanta: mild/moderate/severe (by organ failure).
- **Grey Turner's (flank) + Cullen's (umbilicus) = haemorrhage.**
- **Mild** → **conservative, NO antibiotics.** **Severe** → **ICU + fluids + enteral feeding; ERCP <72h for gallstone + cholangitis.**
- **Necrosis: ANC** → **WON (4 wks); infected necrosis 50% mortality; sterile necrosis don't drain.**
- **Pseudocyst = amylase-rich, fibrous wall, ≥4 weeks; gallstone pancreatitis** → **cholecystectomy same admission.**

PANCREAS (Surgery) — Revision Table

Bold = high-yield. Focus: acute pancreatitis.

ANATOMY & INVESTIGATION

Topic	Key points
Anatomy	Retroperitoneal; head 30%, body+tail 70%; 80–90% exocrine acinar
Islets	75% B (insulin), 20% A (glucagon) , D (somatostatin), PP
Enzymes	Amylase most used (rises hours, falls 4–8d); lipase more sensitive/specific
↑Amylase (non-pancreatitis)	Upper GI perforation, mesenteric infarction, ectopic pregnancy, macroamylasaemia, renal failure, salivary

ACUTE PANCREATITIS — DIAGNOSIS & AETIOLOGY

Topic	Key points
Definition (2 of 3)	Pain + amylase/lipase ≥3× normal + CT findings
Mechanism	Premature enzyme activation → autodigestion → oedema/necrosis
Categories	Mild (interstitial, mortality ~1%); severe (necrotising, 5–10%, mortality 20–50%)
Aetiology	Gallstones 50–70% + alcohol 25%; I GET SMASHED (Idiopathic, Gallstones, Ethanol, Trauma, Steroids, Mumps/malignancy, Autoimmune, Scorpion, Hypercalcaemia/lipidaemia, ERCP, Drugs)
Note	Gallstone cause → cholecystectomy same admission

CLINICAL, SEVERITY & IMAGING

Topic	Key points
Clinical	Pain (epigastric, radiates to back 50%, refractory) + N/V; Grey Turner's (flank)/Cullen's (umbilicus) = haemorrhage ; pleural effusion 10–20%
Severity	Ranson/Glasgow ≥3 at 48h = severe; CRP >150 at 48h ; persisting organ failure; BMI >30. Glasgow = PANCREAS (PaO2, Age>55, Neutrophils, Calcium, Renal urea, Enzymes LDH, Albumin, Sugar)
Atlanta (2013)	Mild (no organ failure); moderate (transient <48h); severe (persistent >48h)
Imaging	Plain film: sentinel loop, colon cut-off, renal halo . US within 24h (gallstones). CECT (Balthazar) if doubt/severe/complications; underestimates necrosis first 72h

MANAGEMENT & COMPLICATIONS

Topic	Key points
Mild	Conservative: IV fluids, NO antibiotics , analgesia/antiemetic, brief fast
Severe	ICU + aggressive fluids + O2 + monitoring ; antibiotics if cholangitis; ERCP <72h (gallstone + cholangitis); enteral (NG) feeding
Systemic complications	Shock, ARDS, renal failure, DIC, hypocalcaemia (first week)
Necrosis	No CT enhancement → ANC → walled-off necrosis (4 wks); infected necrosis = 50% mortality; sterile necrosis NOT drained
Pseudocyst	Amylase-rich, fibrous wall, needs ≥4 weeks ; complications: infection, rupture, haemorrhage (vessel erosion), obstruction
Other local	Abscess, ascites, pleural effusion, portal/splenic vein thrombosis, pseudoaneurysm